PoliTo Automotive Engineering

Dynamic EV Charging Routing & Reservation

Technical Research Questions & Key Performance Indicators

Service Overview

01 Objective

Eliminate **range anxiety** through proactive charger discovery and reservation

02 Actors

EV, Mobility Service Provider (**MSP**) cloud, Charging Point Operator (**CPO**)

03 Data Flow

Continuous **telemetry** (SoC, GPS, consumption) to cloud; downlink with charger options & reservations

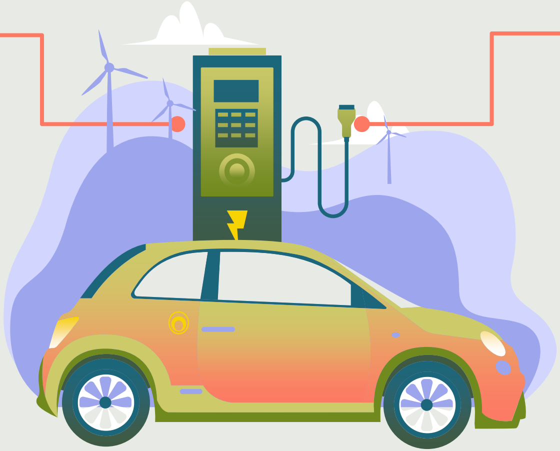
04 Value Prop

Automatic route optimization, guaranteed charger availability, seamless integration with navigation

Two Charging Service Scenarios

Motorway Scenario

- Road: Highway
- Vehicles: BEV
- Traffic: Heavy (2000 veh/h/lane)



Urban Scenario

- Road: City Streets
- Vehicles: BEV
- Traffic: Medium (1000 veh/h/lane)

- Challenge: Range anxiety, charger competition, real-time status, dynamic availability

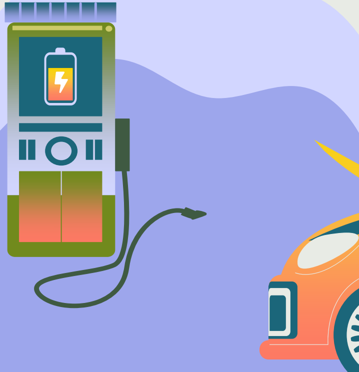
Architecture: In-Vehicle Telematics (TCU) → Cloud Backend (MSP) → CPO Network Integration

Uplink (Vehicle to Cloud):

- SoC (State of Charge)
- GPS Position & Destination
- Consumption Rate (kWh/100km)
- Frequency update: 10-30 seconds

Downlink (Cloud to Vehicle):

- Charger Status & Availability
- Dynamic connector status
- Pricing [€/kWh]
- Route & ETA update
- Latency: < 100 ms



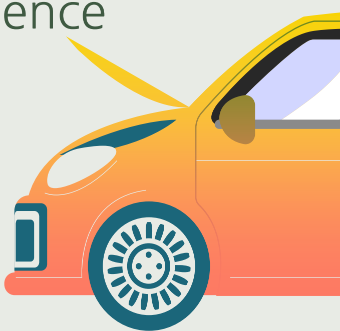
Validation methodology: FESTA framework, FOT (Field Operational Tests), performance indicator tracking.

Research Question:

Is charger status data sufficiently fresh to prevent wasted trips to occupied chargers?

- KPI: **Age of Information (AoI)**
- Target: 30-60 seconds

- @ 130 km/h: 1 km per 28 seconds
- AoI > 2 min = stale data
- Another EV can occupy charger
- 60 sec = user confidence

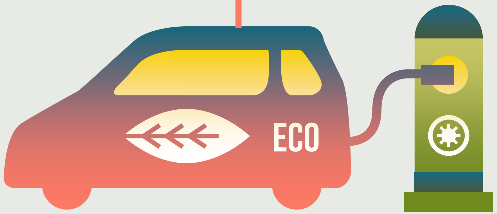


TRQ 2

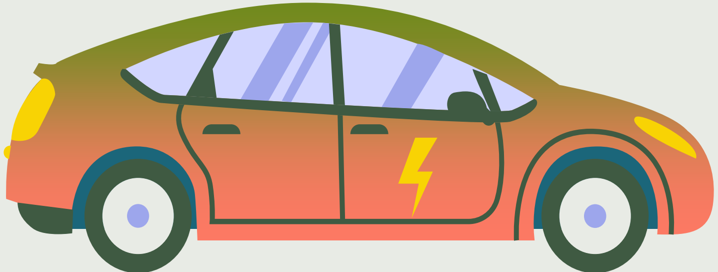
Research Question:

Is connectivity guaranteed along highway routes during the service transaction?

- KPI: **Coverage Probability**
- Target: 99% coverage



- Dead zones (tunnels, remote)
- 99% = 1 km/100 km offline
- @130 km/h = 1 min disconnection every 90 min

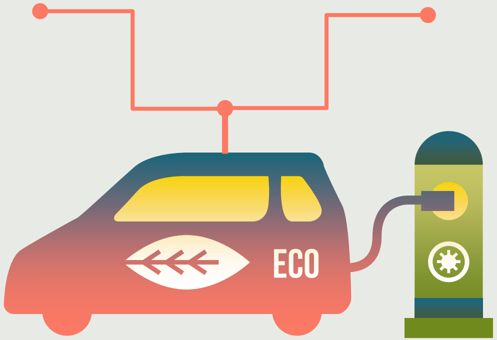


Research Question:

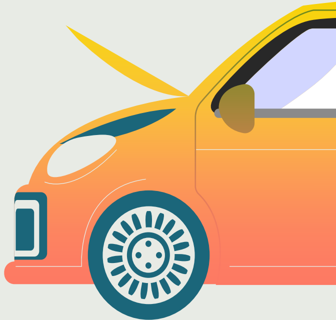
TRQ 3

Does charger reservation complete successfully on first attempt without user distraction?

- KPI: **Service Reliability**
- Success rate: 99%
- Time target: < 5 seconds



- 1 failure in 100 attempts
- Failed reservations = driver distraction
- < 5 sec = no user frustration



Communication Technology Selection

Technology	Coverage	Throughput	Latency	Selected?
4G LTE	99% highway	100 Mbps	30-50 ms	✓ YES
5G NSA	70% highway	1 Gbps	20-40 ms	Future
5G SA	< 10% highway	> 1 Gbps	1 - 10 ms	Future
ITS-G5	N/A (Local)	27 Mbps	100 ms	✗ Unsuitable

Range: 1km

Validation Methodology (FESTA Framework)

1. Preparing

- Define TRQs
- Identify performance indicators
- Recruit test drivers (100-500)

2. Using (FOT)

Field Operational Test

- NFT: Naturalistic Field Trial
- Real traffic conditions
- Continuous data logging

3. Analysing

- Process & storage of FOT data
- Statistical analysis of KPIs
- Answer research questions

4. Results

- Cost-benefit analysis
- Deployment scenarios
- Industry standards

Conclusions & Next Steps

Clear Architecture:

- V2N service with well-defined actors, data flows and transaction

Measurable KPIs:

- AoI (30-60 s), Coverage (99%), Reliability (99% < 5 s) directly address user needs

Technology Ready:

- 4G LTE provides foundation;
- 5G SA computing future optimization

Next Steps:

- Standardization (OCPP 2.0.1), pilot deployment (100-500 vehicles), KPI validation

THANKS!